



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to SUDHARSAN TRADERS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

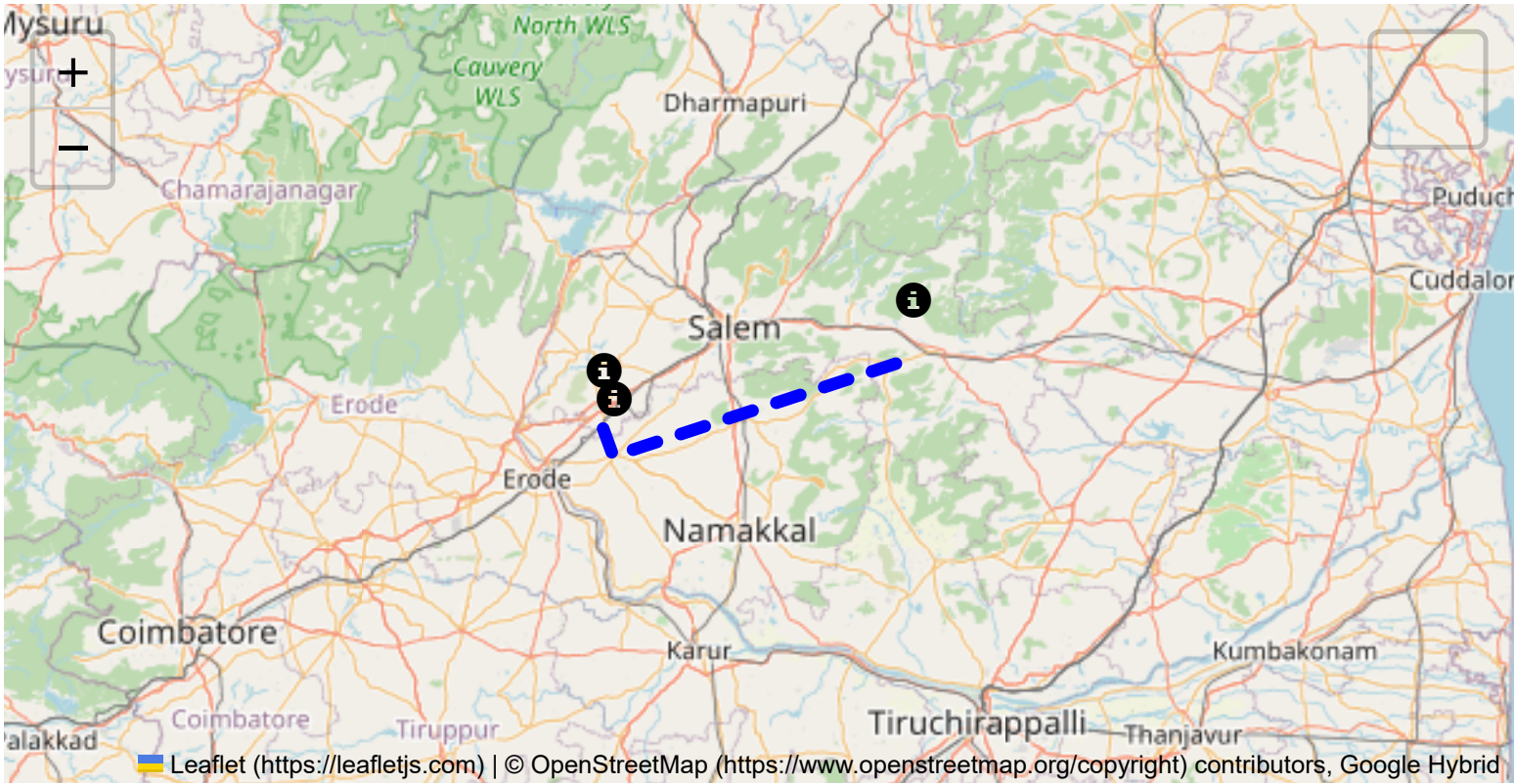
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 92.40 km
Estimated Duration: 2.0 hours
Adjusted Duration (Heavy Vehicle): 2.5 hours
Start: (11.4381, 77.8734)
End: (11.588979, 78.548845)



Welcome to the Journey Risk Management Study

Route Overview

The route from CVQF+23W, Sangagiri to 4, Attur - Rasipuram Rd, through Tiruchengode, spans approximately 92.4 kilometers and involves a mix of state highways and local roads. The journey typically takes around 1.96 hours for a heavy vehicle carrying hazardous materials. Starting from Sangagiri, the

route traverses through Tiruchengode before proceeding to Attur, with a mix of urban and semi-urban landscapes.

Weather Conditions

- **Typical Weather:** The region experiences tropical weather with significant rainfall during the monsoon season (June to September) and high temperatures in summer (March to May).
- **Hazards:** Heavy rainfall can lead to waterlogging and reduced visibility. Summer heat can affect road conditions, leading to the softening of asphalt and tire issues.

Traffic Patterns

- **Peak Hours:** Morning and evening rush hours (7-9 AM and 5-7 PM) in urban areas such as Tiruchengode can see increased congestion.
- **Congestion-Prone Areas:** Notable traffic slowdowns are likely near market areas and intersections within towns, particularly in Tiruchengode.

Road Quality and Infrastructure

- **State Highways:** Generally well-maintained but can have potholes post-monsoon.
- **Local Roads:** Narrow roads with occasional poor patch conditions, especially in rural segments.

Alternative Routes for Emergencies

- **Option 1:** From Sangagiri, head via NH44 and join state highways suitable for heavy vehicles, avoiding congested town centers.
- **Option 2:** Use local bypasses around Tiruchengode for a faster, less congested path, where available.

Local Regulations on Hazardous Materials

- **Transport Permits:** Required for hazardous materials. Compliance with Tamil Nadu Pollution Control Board guidelines is mandatory.
- **Timing Restrictions:** Avoid transporting hazardous materials during festival times or local events when roads are crowded.

Historical Incidents

- **Accidents:** The region has seen occasional heavy vehicle accidents attributed to sharp bends and poor visibility during rains.
- **Materials Spills:** Minor incidents involving chemical spills due to road conditions impacting transport tanks.

Environmental Considerations

- **Sensitive Areas:** Areas near agricultural lands, water bodies between Tiruchengode and Attur should be traversed with caution.

- **Wildlife:** Although not a significant concern, watch for animal crossings in certain rural parts.

Communication Coverage

- **Coverage:** Good in urban and semi-urban areas but potential dead zones in rural stretches between Sangagiri and Attur.
- **Alternatives:** Reliance on offline maps and direct radio communication may be necessary in low connectivity areas.

Emergency Response Times

- **Urban Segments:** Tiruchengode offers quicker response times due to proximity of emergency services.
- **Rural Segments:** Longer response times, potentially over an hour in case of accidents or spills.

Overall Risk Assessment Summary

- **Risks Identified:** Weather-related challenges, road quality, traffic congestion, and communication gaps.
- **Mitigation Suggestions:** Travel during non-peak hours, regular weather updates, adherence to local safety regulations, and emergency preparedness with alternative routes.
- **Conclusion:** With careful planning and consideration of the noted challenges, the route is manageable for transporting hazardous materials, although the risks necessitate vigilance, especially in unpredictable weather or road conditions.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	Turn	High	11.43968, 77.87345	15 KM/Hr
2	Turn	High	11.44019, 77.87547	15 KM/Hr
3	Turn	Medium	11.39685, 77.88861	30 KM/Hr
4	Turn	High	11.37868, 77.89498	15 KM/Hr
5	Turn	Medium	11.37809, 77.89834	30 KM/Hr
6	Turn	Medium	11.37885, 77.89989	30 KM/Hr
7	Turn	Medium	11.37820, 77.90220	30 KM/Hr
8	Turn	Medium	11.36748, 77.93938	30 KM/Hr
9	Turn	Medium	11.38968, 78.02925	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
10	Turn	Medium	11.41310, 78.08062	30 KM/Hr
11	Turn	High	11.41468, 78.08072	15 KM/Hr
12	Turn	High	11.42964, 78.11587	15 KM/Hr
13	Turn	Medium	11.42447, 78.15264	30 KM/Hr
14	Turn	Medium	11.42962, 78.16871	30 KM/Hr
15	Turn	Medium	11.43142, 78.18781	30 KM/Hr
16	Turn	High	11.43542, 78.20754	15 KM/Hr
17	Blind Spot	Blind Spot	11.45099, 78.20397	10 KM/Hr
18	Turn	Medium	11.44993, 78.22146	30 KM/Hr
19	Turn	Medium	11.44956, 78.22253	30 KM/Hr
20	Turn	Medium	11.44909, 78.22293	30 KM/Hr
21	Turn	High	11.44800, 78.22312	15 KM/Hr
22	Turn	Medium	11.44758, 78.22397	30 KM/Hr
23	Turn	Medium	11.49825, 78.32406	30 KM/Hr
24	Turn	Medium	11.50502, 78.33084	30 KM/Hr
25	Turn	Medium	11.50779, 78.33309	30 KM/Hr
26	Turn	Medium	11.50863, 78.33962	30 KM/Hr
27	Turn	Medium	11.51862, 78.34812	30 KM/Hr
28	Turn	Medium	11.54826, 78.35614	30 KM/Hr
29	Turn	Medium	11.56714, 78.37807	30 KM/Hr
30	Turn	Medium	11.57102, 78.50078	30 KM/Hr
31	Turn	Medium	11.58683, 78.54570	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	hospital	T.C.A Hospital Tiruchengode	11.3791885, 77.8965774	30 km/h	Medium
6	clinic	Kongu Nursing Home	11.3783065, 77.8961134	30 km/h	Medium
7	hospital	Soorya Multispecialty Hospital	11.3786429, 77.8931912	30 km/h	Medium
8	hospital	Tiruchengode Government Hospital	11.37645, 77.89426	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
13	hospital	Government Hospital, Namagiripettai	11.4569484, 78.265263	30 km/h	Medium
14	clinic	Dr. Saravanakumar Clinic	11.5842002, 78.4441223	30 km/h	Medium
15	hospital	Government Hospital	11.572472, 78.497407	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium
9	marketplace	Farmer Market	11.4146849, 78.0805463	30 km/h	Medium
10	school	Government Higher Secondary School	11.4150045, 78.0838172	30 km/h	Medium
11	school	Government school	11.4248027, 78.1323902	30 km/h	Medium
12	college	Paavai Arts And Science College For Women	11.4262443, 78.1546987	30 km/h	Medium
16	school	Rasi Matric HR Sec School	11.5760027, 78.5062808	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.38395, 77.89480



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44019, 77.87547



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.39685, 77.88861



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.37868, 77.89498



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.37809, 77.89834



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.37885, 77.89989



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.37820, 77.90220



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.36748, 77.93938



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.38968, 78.02925



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41310, 78.08062



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.41468, 78.08072



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.42964, 78.11587



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.42447, 78.15264



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.42962, 78.16871



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.43142, 78.18781



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.43542, 78.20754



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.45099, 78.20397



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44993, 78.22146



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44956, 78.22253



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44909, 78.22293



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44800, 78.22312



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44758, 78.22397



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49825, 78.32406



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.50502, 78.33084



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.50779, 78.33309



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.50863, 78.33962



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.51862, 78.34812



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.54826, 78.35614



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.56714, 78.37807



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.57102, 78.50078



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.58683, 78.54570